Who Must Consent to an Incentive Sale?

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Abstract

An artist relies on a manager to develop her career and on a financier to fund it. The manager is paid a commission on the artist's income, both from recorded works, which do not require her later effort, and from activities that do, such as touring. The financier can buy that commission without changing the artist's share. The manager then has no stake in the income that depends on the artist and develops only recorded works; the artist bears the loss. Anticipating such a sale, less of the development that depends on the artist can be financed. We price the artist's right to veto the sale. The right lets her projects combine recorded works with income from her own effort, but where development costs rise faster than quadratically the manager must be paid a rent to develop them. For small costs, the price of consent is the resulting cost saving net of that rent. The results have three implications for contract design. First, the artist's payment share and her authority over the manager's commission are distinct terms, and bargaining over the share alone leaves the authority unpriced. Second, when costs rise at most quadratically, the right is worth most to the artists with the least wealth. Third, when costs rise faster, artists above a wealth threshold should pair a protected manager with projects under the financier's direct control. The right is valuable, and it moves the boundary between delegated and integrated development, although the sale never changes her share.

JEL: D86, G32, L14, Z11.

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1 Introduction

A music manager is typically paid a commission on the artist's earnings, both from recorded works and from touring, endorsements, and other activities that require the artist's continued effort. The commission is itself an asset that a financier can buy, and contracts allocate control over such a sale in opposite ways. A specimen agreement published by British artist, manager, and musician organizations bars the manager from assigning "the benefit" of the agreement without the artist's consent, which she may withhold at her absolute discretion; the standard exclusive contract issued by Korea's Ministry of Culture, Sports and Tourism requires the artist's prior written consent before the agency transfers any part of its contractual rights; a management agreement filed with the SEC lets the manager assign its rights "in whole or in part" after notice and consultation (Section 7). Why should the artist care about a sale that leaves her own payment share unchanged, and what is a right to block it worth? This paper answers both questions and shows how the answer depends on the alternative of letting the financier direct development itself.

The manager directs the artist's career, and the commission ties that direction to income that depends on the artist's later effort. After a sale, the manager still directs the career and bears the cost of developing it, but no longer shares in that income, so development shifts toward recorded works and the artist bears the cost. The sale is public and anticipated when the initial contract is written, and its cost falls on a party who is not part of the sale and must exert effort afterwards. Nothing the two traders promise each other can protect her, because they can revise any such promise together whenever the sale is profitable. Nor can her payment share: it affects the manager's incentives only through the manager's contingent pay, and after the sale the manager holds none. Protection must take the form of authority over the sale itself. This is a different problem from the one studied in the literature on claim sales. When a bank sells a loan or an originator sells cash flows, the incentive that weakens is the seller's, its cost is reflected in the price the buyer pays, and the question is how much of the claim the seller should retain (Pennacchi 1988; Gorton and Pennacchi 1995; Parlour and Plantin 2008; Vanasco 2017). In sequential and nonexclusive contracting, a later contract weakens the incentives of the party who signs it (Bizer and DeMarzo 1992; Kahn and Mookherjee 1998). Here the question shifts from how much of an incentive claim to retain to who must consent to its sale.

The relevant organizational alternative is integration: the financier directs a project's development itself rather than delegating it to a manager. Integration avoids agency rents, but it faces the same limit on pledgeable income. Without consent, the holders of contractible claims can always fall back on developing only recorded works, through a buyout

of the manager under delegation or through the financier's own choice under integration. The recording income of a project with any artist-dependent development is therefore their outside option and cannot be pledged. Such a project gains most by specializing fully in artist-dependent income, while recording income is pledged by projects that specialize in recorded works. Financing without consent thus relies on a portfolio of specialized projects. Consent makes delegation credible and permits balanced projects, but the manager must be paid enough to implement them. Depending on development technology, limited liability may leave the manager a rent. The value of consent is the saving from replacing specialized projects with balanced ones, net of any such rent. Consent therefore moves the boundary between delegated and integrated development: the optimal organization selects the lowest-cost combination of integration and protected delegation consistent with financing. The same governance problem arises whenever a developer's incentive pay can be sold while another party's later effort determines part of the income. An engineer can orient a start-up toward assets that can be sold or licensed, or toward a pipeline that depends on the founder's continued work; a lead arranger continues to administer a syndicated loan while selling its retained share. We develop the artist case and return to these settings in Section 7.

The model has three parties: a creator A, the artist; a developer D, her manager; and a competitive financier F. Development allocates a project between work income, from recorded works and other assets whose receipts do not require A's later effort, and personal income, from touring, endorsements, and other activities that do. Development is not contractible, and payments are limited by realized cash. Without consent, F can buy D's contingent claims after contracting. The buyout needs only a sale of payment rights, not a contract on the hidden development choice. Its anticipation gives F and D a joint outside option equal to their receipts from work-only development, which limits the income that can be pledged to finance personal development. F can instead integrate development and choose the direction itself; this removes the trade together with delegated incentives but leaves the same limit on pledgeable income. With consent, a reallocation of D's incentive pay also requires A's agreement. Initial financing can cover a portfolio of projects, such as several of an artist's projects funded under one agreement, in which work-oriented projects supply resources to personal-development projects and organizations can differ across projects.

The main theorem makes this precise in a joint limit in which contractible personal income and development costs are both small. Let λ denote the mean development direction that financing supports, which rises with the artist's wealth, and write normalized development cost as $g(t)$. A protected manager can be motivated to choose direction t at expected pay

$$\Phi(t) = \max\{g(t),\, tg'(t)/2\}$$

to first order. The first term is development cost, which participation requires. The second is the pay needed when the most informative payment state has a probability growing at rate $2/t$: that state combines personal success with work failure, so its probability responds to development through both margins. A single contract attains the larger of the two. Where the incentive bound is higher, it uses a flat personal claim and leaves the manager a rent. Where participation binds, the claim on personal income rises with the direction, so a higher direction lowers A's effort and slows the growth of the manager's bonus probability. The artist's own effort response thus disciplines the manager, and development is implemented at cost. The relevant frontier $\mathcal{E}_0$ is the lower convex envelope of Φ and the two integrated endpoints. The first-order price of consent is

$$\lambda g(1) - \mathcal{E}_0(\lambda),$$

the distance between the cost of the unprotected integrated portfolio and this envelope. The theorem covers all cash-feasible payment schedules and optimizes capacity and income coverage. It applies to general development technologies under two conditions, at a sufficiently small cost level; both hold for every power cost at least quadratic, and for subquadratic power costs under an explicit cash-slack condition.

The envelope describes an organizational transition when implementation requires rents. At lower wealth, a single protected manager chooses an intermediate direction. At higher wealth, the optimal portfolio combines a protected intermediate project with integrated personal development: integration economizes on rents where commitment is not needed. On this mixed segment, greater wealth lowers the price of consent. With a single protected manager, the price rises with wealth while the marginal implementation cost $\Phi'(\lambda)$ is below the integrated endpoint cost $g(1)$ and falls afterwards. The wealth effect is therefore a property of development technology.

The incentive bound exceeds development cost exactly where the elasticity of development cost exceeds two, a cutoff that reflects the $2/t$ growth rate of the informative state in this technology. Quadratic costs sit at the boundary. For them $tg'(t)/2 = g(t)$, so implementation rents vanish, mixed governance does not arise, and the price reduces to a convexity saving. In this case we solve the finite-parameter problem exactly. With $h(y) = \chi y^2/2$, $\nu = w/v^2$, and contractible income and χ shrinking proportionally,

$$d^*(w) = \frac{\chi v^2}{2}\left(\frac{1}{4} - \nu^2\right) + O(\alpha^2),$$

the saving from developing one balanced project rather than a portfolio of specialized ones.

Over a continuous parameter region, exact certificates show that strictly implementing protected contracts finance fees above 4% of the optimized unprotected value. If protection is paid for only in projects that use a protected manager, the highest first-order fee per protected project is $\max_t\{tg(1) - \Phi(t)\}$, independent of wealth.

For contract design, the results have three implications. An artist's payment share and her authority over the manager's commission are separate terms: the share reaches the manager only through the commission, which a sale removes, while the authority keeps the commission in place, so negotiating over shares alone misses the value of consent. When development costs rise at most quadratically, consent is worth most to artists who depend most on outside financing. When power costs rise faster than quadratically, an artist above a wealth threshold does best by delegating balanced projects to a protected manager and placing projects devoted entirely to artist-dependent income under the financier's direct control. Section 7 relates these implications to existing management agreements.

The results differ in scope. The buyout bound, the contract that implements a protected manager's direction at expected pay Φ with rents exactly where cost elasticity exceeds two, and its strict and stable versions concern a single contract and a single project. The price of consent, the mixture of delegation and integration, the wealth effects, the exact quadratic price, and both fee conventions compare portfolios, and they apply where financing can be pooled across projects.

In the claim-sale literature, Vanasco (2017) shows that costly retention of cash flows is needed to implement an originator's screening when those cash flows can later be sold, and Gryglewicz, Mayer, and Morellec (2024) analyze the dynamics of the monitoring trade-off in loan sales. These analyses, like the sequential contracting of Bizer and DeMarzo (1992) and Kahn and Mookherjee (1998), concern the incentives and payoffs of the parties to the sale or to the subsequent contract. Here the claim is traded between F and D, while the productive party harmed by the resulting change in development is A, who is outside the trade. The financing consequence builds on the distinction between total surplus and pledgeable income in Holmström and Tirole (1997) and Tirole (2006).

Bilateral contracts that affect third parties are studied by Segal (1999), Segal and Whinston (2003), and Baliga and Sjöström (2009), and side contracting is central to Tirole (1986), Laffont and Martimort (1998), and Dessí (2005). Demski and Sappington (1991) use buyouts to resolve double moral hazard; here a buyout destroys the incentives another productive party needs. Three features distinguish the present problem: the harmed third party remains productive after the trade, the remedy is a veto over claims she does not own, and the value of that veto depends on the endogenous alternative of integrating development. Hart and Moore (1988), Dewatripont (1989), and Fudenberg and Tirole (1990) study how

renegotiation shapes initial incentives; we construct strictly implementing contracts that are immune to unanimous Pareto-improving amendments. The competing development directions connect to Holmström and Milgrom (1991): a high-powered incentive can correct task allocation, but its subsequent sale removes that correction.

Aghion and Bolton (1992) connect financial contracting to contingent control, and Aghion, Bolton, and Tirole (2004) study liquidity, incentives, and exit rights, including the allocation of veto authority. The ownership approach of Grossman and Hart (1986) and Hart and Moore (1990) allocates residual control through assets; consent instead protects another party's incentive exposure without transferring ownership to the artist. Bai, Tao, and Wu (2004) relate revenue shares to control rights in team production; we separate an artist's payment share from authority over her manager's claims. Meyersson Milgrom, Milgrom, and Singh (2023) explain why stakeholders often hold targeted veto rights over specific classes of decisions while management retains broad control; here the veto covers transfers of incentive claims, and its value is derived from the financing it makes possible. Ayotte and Bolton (2011) study when rights created in financial contracts should bind third parties. Here the third party holds authority over the transfer itself, and the value of that authority comes from the organizational arrangements it makes feasible.

2 Environment and organizational choices

2.1 Production, effort, and cash

There are three risk-neutral parties: creator A, developer D, and financier F. In the leading application, A is an artist, D is her manager, and F funds the career and can buy the manager's commission. A has wealth $w \geq 0$. The initial contract chooses a project's capacity $p \geq 0$ and additional income coverage $\theta \in [0, 1]$, defined below, at cost

$$C_0(p, \theta) = \frac{p^2}{2} + \kappa p\theta, \qquad \kappa > 0.$$

D subsequently chooses a noncontractible development direction $t \in [0, 1]$ and incurs cost $h(pt)$, where $h(0) = 0$ and $h(y) > 0$ for $y > 0$. Higher t shifts development toward income that requires A's participation, such as touring and endorsements, and away from recorded works. Work income is pv, with success probability $1 - t$ and $v > 0$.

A personal-income opportunity arrives with probability $\pi \in (0, 1]$. Put $R = r + a$, with $r, a > 0$. Conditional on an opportunity, personal income is pR with probability $t(r/R + e_A)$, where r/R is baseline success and e_A is A's effort contribution. A observes t and chooses

effort $e_A \in [0, a/R]$, at cost $ptR^2e_A^2/(2a)$. Work and personal success are conditionally independent. A single direction therefore moves two independent success margins: higher t lowers the chance of work success and raises the chance of personal success. If no one else holds a claim on personal income, A chooses $e_A = a/R$, and expected personal resources net of effort cost are $pt(r + a/2)$.

A fraction $1 - \mu$ of personal opportunities is contractible at baseline, where $\mu \in [0, 1]$, and coverage θ makes a fraction θ of the remaining μ contractible. We call contractible opportunities covered. For an artist, coverage determines how much touring and endorsement income the agreement captures. Coverage is chosen in the initial contract so that the price of consent does not rest on a fixed contractible share of personal income. Thus

$$\omega = \pi(1 - \mu + \mu\theta), \qquad \epsilon = \pi(1 - \mu), \qquad \gamma = \frac{a}{R^2}, \qquad q = \frac{\pi(r + a/2)}{v} - 1.$$

Here ω is the probability of a covered personal opportunity and ϵ its baseline value at $\theta = 0$. The parameter q measures the social advantage of personal development relative to work. Uncovered personal income accrues to A and induces first-best effort. Throughout the main results $q > 0$.

Payments depend on realized contractible cash. When there is no covered personal opportunity, state N denotes work success. With a covered opportunity, states 10, 01, and 11 denote work only, personal only, and joint success; the first digit records work success and the second personal success. Zero-cash states pay nothing. A commission proportional to income is one such schedule; the analysis allows any cash-feasible state-contingent schedule, so the value of consent does not depend on a particular commission form. Total claims held by F and D, divided by p, are $g_N, g_{10}, g_{01}, g_{11}$, bounded respectively by $v, v, R, v + R$. F and D receive nonnegative shares of these claims. The contract may also specify nonnegative initial advances, fixed payments made at the outset and financed together with the project. A receives the residual cash and may discard cash before settlement, so her retained income cannot fall when more cash is realized; among equally good retention choices, A retains the most actual cash. Appendix A incorporates this settlement option explicitly.

2.2 Trades, consent, and integration

An initial contract specifies capacity, coverage, payment schedules, and an organization. Under delegation, D chooses development and bears its cost. After contracting but before development, F can purchase all of D's contingent claims for a fixed payment. A's total payment schedule remains unchanged, and D remains responsible for development. Once bought out, D has sold all contingent pay but still bears development cost, so she chooses

$t = 0$. An unprotected delegated contract must therefore be stable against this buyout.

Under integration, F chooses development and incurs its cost. F faces the same development cost as D, so integration and delegation differ in who decides and who holds incentive pay, not in development skill. Integration allocates the development decision for a project, not the career-level management relationship: an artist can retain a manager for her career while a financier controls the development of particular projects. Let $\mathcal{U}$ denote the organizations available without consent: integration and buyout-stable delegation. Under protected delegation, denoted $\mathcal{P}$, A's consent is required for a reallocation of D's incentive compensation. Consent changes who must agree to such a trade: one that leaves F and D better off at A's expense cannot proceed without A. The right covers economically equivalent reallocations of D's contingent pay, including side transactions, rather than only amendments to the face of the original agreement.

To convexify the organizational menu, the initial contract can specify a public lottery, resolved before construction and development. Financing is committed before resolution: F's break-even condition holds in expectation across branches, so financing is pooled, while participation, effort incentives, and postcontract stability hold in each realized branch. A lottery can equivalently represent a financed portfolio: work-oriented projects supply resources to personal-development projects. The lottery is a convexification device for jointly financed projects, not a claim that a single project is literally randomized across organizational forms. We allow portfolios of organizations in $\mathcal{U}$ and also study the expanded menu

$$\mathcal{M} = \mathrm{lot}(\mathcal{U} \cup \mathcal{P}).$$

This permits a governance arrangement to use delegation in some projects and integration in others. Pure protected delegation remains feasible throughout. Results stated for a single contract hold in each realized branch; the price results compare portfolios.

2.3 Financing and fees

A proposes the initial agreement. Competitive financiers break even in expectation, and D's reservation payoff is zero. Let C_D be expected developer compensation and $R_D = C_D - h(pt)$ its rent. Write S for the combined resources of A and F, net of construction and developer compensation. Write Q for F's receipts net of the same expenditures, before the protection fee and A's wealth contribution. Under integration, developer compensation is replaced by development cost and $R_D = 0$. A's payoff gross of contributed wealth is $S - Q$; we call this the creator's continuation payoff.

A can contribute at most her wealth, so financing requires $\mathbb{E}Q + w \geq 0$. For an unpro-

tected arrangement,

$$V_U(w) = \sup_{\text{lot}(\mathcal{U})} \{\mathbb{E}S : \mathbb{E}Q \geq -w\}. \tag{1}$$

The main analysis prices a governance package that costs d in every realized branch. The cost is incurred before development, so it both uses resources and must itself be financed:

$$V_M(d, w) = \sup_{\mathcal{M}} \{\mathbb{E}S : \mathbb{E}Q \geq d - w\} - d. \tag{2}$$

The package price is the largest d for which $V_M(d, w) \geq V_U(w)$: the most A would pay for protection when contracts, financing, and organization are chosen optimally, rather than a price set by bargaining over the right. Charging only branches that use protected delegation is a separate convention, analyzed in Section 6.

No new information arrives between contracting and development. Consent therefore does not block any information-driven efficient revision in this environment, and the fee d is the resource cost of protection. The initial contract fully anticipates the trade, so the problem is not incomplete foresight. A clause in which F and D promise not to trade can be revised by F and D alone, and they gain from revising it whenever the buyout is profitable (Section 3). Consent changes who must join that revision: under $\mathcal{P}$, a reallocation of D's incentive pay requires the agreement of A, who bears its cost. Figure 1 summarizes the timing. Claim trades and amendments occur after the organization is realized and before the development direction is chosen; the payment schedule then governs effort and settlement. At that stage, within the realized capacity, coverage, and decision technology, parties may rewrite cash-feasible payments and arrange fixed compensation. We call an agreement *renegotiation-proof* if no such deterministic or randomized amendment weakly benefits all participants and strictly benefits at least one. Weak incentive compatibility defines the closed feasible set for optimization. Where a limiting contract leaves D indifferent, we establish the same value as a supremum of strictly implementing contracts, which can also be chosen renegotiation-proof under the conditions of Proposition 2. Development itself remains noncontractible throughout.

A trade after development would be neutral under additional conditions. Suppose the realized direction is known to F and D and A's payment schedule is unchanged. A voluntary reallocation of claims between F and D then leaves A's effort and the cash distribution unchanged. With development costs sunk, their combined conditional payoff is fixed, so both voluntary-acceptance constraints must bind and each keeps her conditional expected payoff. If the same trading opportunity is available after every observable direction, development incentives are therefore unaffected. The consequential trade is the one before development.

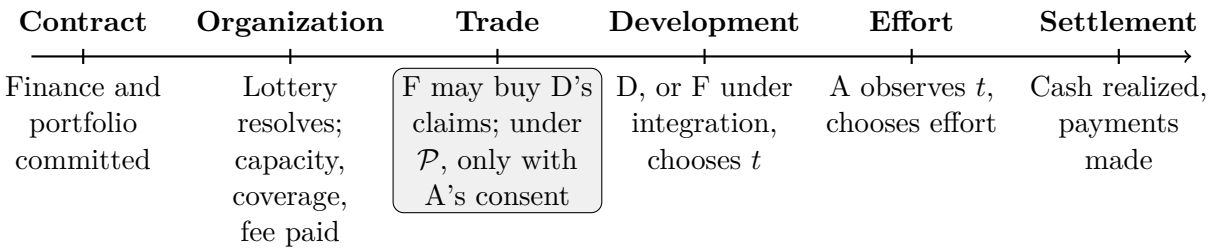


Figure 1: Timing.

3 Tradable incentives and pledgeable income

Let $J(t)$ be the joint continuation payoff of F and D, net of development cost, and let pG be their joint receipts from work-only development after a buyout. The sale reallocates income between F and D without changing A's total payment schedule.

Proposition 1 (Buyout bound)**.** *Every buyout-stable delegated contract satisfies $J(t) \geq pG$. For a positive development direction, this condition bounds F's receipts net of fixed pay by personal-state operating profit, less $h(pt)/t$ and D's rent. The same bound applies under integration.*

Proof. After buying all contingent pay, F and D jointly receive pG, since D uniquely chooses zero development. If $J(t) < pG$, a fixed price can make both parties strictly better off. Under integration, work-only development is an available deviation. Appendix A gives the resulting financing inequality. □

The bound uses an ordinary transaction in payment rights. It does not give F and D a contract on the hidden action. Initial incentive compatibility and stability against the later sale are therefore different restrictions. A contract satisfying the former can promise a large personal-success bonus and finance itself with work receipts. The possibility of a buyout makes those receipts an outside option for F and D, the holders of the contractible claims, which restricts their pledgeability before the trade occurs.

The argument is unchanged if amendments are randomly selected or if a bargaining procedure gives one party all the gains from trade, provided a mutually beneficial buyout can be proposed and accepted. A bilateral promise by F and D never to trade is itself subject to a mutually beneficial revision. Consent places the veto with the party harmed by the trade. The unprotected optimum derived below is attained by integration, so richer F–D bargaining that retains the available buyout does not improve the unprotected frontier.

An accounting reduction is useful for the remaining analysis. Appendix A shows that, after eliminating disposal, personal claims can be written $m(u) = c + bu$. Work claims are

$g_N = v$ and $g_{10} = v + \min\{0, b\}$, while $g_{01} = c + b$ and $g_{11} = g_{10} + c$. Put $m_+ = \max\{m, 0\}$. At the implemented direction,

$$S = p[v + vqt - \omega\gamma t m_+^2/2] - C_0 - C_D, \tag{3}$$

$$Q = p[G(1-t) + \omega t m(1 - \gamma m_+)] - C_0 - C_D, \tag{4}$$

where $G = v + \omega \min\{0, b\}$. A negative personal-claim slope can reduce the compensation needed to motivate D, but it also reduces work receipts available to F. The main theorem accounts for both effects.

4 Implementation cost and the price of consent

4.1 A small-coverage, small-cost economy

Hold the income technology fixed and consider

$$\epsilon = \alpha\gamma v, \qquad h_\alpha(y) = \alpha H(y), \qquad g(t) = \frac{H(vt)}{v^2}, \qquad \alpha \downarrow 0. \tag{5}$$

Here g without a state subscript is normalized development cost. The limit keeps the cost of motivating development comparable to the covered cash available to pay for it. The coverage share is recovered as $\mu = 1 - \alpha\gamma v/\pi$. Define

$$\nu = \frac{w}{v^2}, \quad \lambda = \frac{1}{2} + \nu, \quad x = \frac{q}{1+2q}, \quad F(q) = \frac{q^2}{2(1+2q)}, \quad K(t) = \frac{t g'(t)}{2}. \tag{6}$$

At zero coverage and cost, $\omega = 0$, $C_D = 0$, and capacity is v. Equations (3) and (4) then give

$$\frac{S_0(t)}{v^2} = \frac{1}{2} + qt, \qquad \frac{Q_0(t)}{v^2} = \frac{1}{2} - t,$$

so the financing constraint $\mathbb{E}Q_0 + w \geq 0$ is $\mathbb{E}t \leq \lambda$. Since $q > 0$, the mean direction equals λ. Wealth therefore enters through λ, and the remaining question is the cheapest way to implement that mean direction. An endpoint portfolio and an intermediate protected project have the same limiting value $V_0 = v^2(1/2 + q\lambda)$. Their first-order financing and implementation costs select the optimal organization.

Put $\Delta = r/R$, $\Gamma = \gamma R = 1 - \Delta$, $E = \gamma v$, and $k = 1 - x$. Write $g = \beta g_1$, where the shape g_1 is fixed and $\beta > 0$ is the level of development cost. The protected implementation cost is

$$\Phi(t) = \max\{g(t), K(t)\}, \qquad \psi_t = 2 - \frac{t g'(t)}{\Phi(t)} \in [0, 1]. \tag{7}$$

The first term is the developer's participation requirement. The second is the pay required when the most informative payment state has a probability growing at rate $2/t$, as shown below. Since $K \geq g$ exactly when the elasticity tg'/g is at least two, $\Phi = g$ where that elasticity is at most two and $\Phi = K$ where it is at least two. If $\Phi(1) > g(1)$, let $t_0 \in (0,1)$ solve

$$\Phi(t_0) + (1 - t_0)\Phi'(t_0) = g(1); \tag{8}$$

otherwise put $t_0 = 1$. The tangent to Φ at t_0 passes through the integrated personal endpoint $(1, g(1))$. Fix $\bar{\nu} < 1/2$, put $\bar{\lambda} = 1/2 + \bar{\nu}$, and let $\mathcal{A} = [\min\{1/2, t_0\}, \min\{\bar{\lambda}, t_0\}]$ denote the protected directions relevant for package fees.

The index ψ_t determines the slope of the personal claim $c + bu$ in the contract that implements t (Section 4.2). That contract pays D only in state 01, whose probability grows at rate $2/t$ when the personal claim is flat. A claim that rises with the direction lowers A's effort at higher directions and slows this growth to $2/t - \gamma b/k$. With expected bonus $\Phi(t)$, D's marginal return at the target is therefore $\Phi(t)(2/t - \gamma b/k)$, and equating it to $g'(t)$ gives $\gamma b = k\psi_t/t$. Where the incentive bound binds, $\psi_t = 0$ and the claim is flat; where participation binds, $\psi_t > 0$. Condition (iii) of the following assumption is the global incentive constraint of this contract, and condition (iv) keeps its claims within cash.

Assumption 1. *(i)* The income technology is feasible, $q > 0$, $x < \Gamma$, and $\kappa > \pi q/[4\gamma(1+q)]$. *(ii)* The function g is increasing and strictly convex with $g(0) = g'(0) = 0$; H is continuously differentiable on bounded intervals and twice continuously differentiable away from zero; Φ is strictly convex and twice continuously differentiable on $(0,1]$, with $\Phi'(t) \to 0$ as $t \downarrow 0$. *(iii)* For every $t \in \mathcal{A}$ and $u \in [0,1]$,

$$g(u) - g(t) \geq \Phi(t)\left\{\left(\frac{u}{t}\right)^2\left[1 + \psi_t - \psi_t\frac{u}{t}\right] - 1\right\}. \tag{9}$$

(iv) For every $t \in \mathcal{A}$,

$$k\psi_t < x + E, \qquad k\psi_t\frac{1-t}{t} < \Gamma - x. \tag{10}$$

Conditions (i), (iii), and (iv), and the cutoff t_0, do not depend on β. Inequality (9) is the global incentive constraint of the contract constructed below; for $\psi_t = 0$ it reads $g(u) - g(t) \geq K(t)[(u/t)^2 - 1]$. Condition (10) keeps that contract's claims within cash. The theorems hold for every sufficiently small level β. Appendix B gives the explicit bound, which leaves cash for the bonus, strictly implements the integrated personal endpoint, and makes the financing cost of a negative claim slope exceed its incentive benefit. The condition on κ makes expansion of coverage unattractive at the relevant financing prices.

For power costs $g(t) = \beta t^n/n$ with $n > 1$,

$$\Phi(t) = \beta \max\{1/n, 1/2\}\, t^n,$$

and (9) holds for every $n > 1$ (Appendix B.5). For $n \geq 2$, $\psi_t = 0$ and (10) holds automatically, so the theorems cover every such power cost at a sufficiently small level. For $1 < n < 2$, $\Phi = g$, $t_0 = 1$, $\psi_t = 2 - n$, and (10) becomes

$$(2 - n)(1 - x) < \min\{x + E,\ \Gamma - x\}.$$

With $v = r = a = 1$ and $q = .4$, this holds for every $n > 23/14$.

4.2 Implementation costs and the envelope

Fix a target $t \in \mathcal{A}$, capacity v, and baseline coverage. Use total claims $(g_N, g_{10}, g_{01}, g_{11}) = (v, v, c + b, v + c)$ with

$$\gamma b = \frac{k\psi_t}{t}, \qquad \gamma c = x - k\psi_t,$$

so that the personal claim at the target is $\gamma m(t) = x$. Pay D only in state 01, with normalized bonus

$$z = \frac{\Phi(t)}{\gamma k t^2}. \tag{11}$$

At a deviation u, the effective claim $c + bu$ gives personal-success intensity $\min\{1, k + \gamma b(t - u)\}$. State 01 has probability $\alpha\gamma v\, u^2$ times this intensity. Work failure and personal success each contribute a factor u, so with a flat claim this probability grows at rate $2/u$, the rate in the definition of Φ. Hence D's payoff divided by αv^2 is at most

$$\Phi(t) \left(\frac{u}{t}\right)^2 \left[1 + \psi_t - \psi_t \frac{u}{t}\right] - g(u),$$

with equality at $u = t$. Inequality (9) is therefore the global incentive constraint. Condition (10) and a small cost level keep all claims and the bonus within cash. Expected compensation is $\alpha v^2 \Phi(t)$, and D's rent is $\alpha v^2[\Phi(t) - g(t)]$.

The two terms of Φ correspond to two regimes. Where the elasticity of development cost exceeds two, incentive provision binds: $\psi_t = 0$, the personal claim is flat, and D earns a rent. State 01 is informative in two ways: personal success favors personal development, while work failure also favors a high personal direction. Limited liability prevents D from paying for that informativeness. Where the elasticity is below two, participation binds. A flat-claim bonus that covers development cost would give D a positive marginal return at the

target, and she would overshoot. The rising claim $b > 0$ corrects this through the creator: a higher direction raises the total claim on personal income and lowers A's effort, slowing the growth of D's bonus probability. The creator's effort response disciplines the developer, and development is implemented at cost. Quadratic cost, with elasticity two, is the boundary at which the two constraints coincide.

Raising the bonus to $\Phi(t) + s$ for small $s > 0$, while keeping the target claim at x and replacing $\Phi(t)$ by $\Phi(t) + s$ in ψ_t, adds

$$s\left(1 - \frac{u}{t}\right)^2\left(1 + \frac{2u}{t}\right) > 0 \qquad (u \neq t) \tag{12}$$

to D's incentive margin against every deviation. Every construction below therefore has strictly implementing approximations at an $o(\alpha)$ cost.

Minimizing the developer's compensation is not the same as maximizing pledgeable income, so Φ need not be the lowest wage across every claim schedule. A declining personal claim raises the growth rate of the bonus probability and can reduce the wage below Φ, but it also reduces the work receipts available to F. The theorem below shows that this financing loss prevents such a claim from lowering the relevant envelope (Lemma B.1). This is the step that connects the geometry to the unrestricted contract problem.

Let $\mathcal{E}_0$ be the lower convex envelope of Φ and the integrated endpoints $(0, 0)$ and $(1, g(1))$. Strict convexity gives

$$\mathcal{E}_0(t) = \begin{cases} \Phi(t), & 0 \leq t \leq t_0, \\ \Phi(t_0) + \Phi'(t_0)(t - t_0), & t_0 \leq t \leq 1. \end{cases} \tag{13}$$

When $t_0 = 1$, the envelope is Φ itself. Figure 2 illustrates the envelope for cubic cost.

Theorem 1 (The price of a governance package)**.** *Under Assumption 1, there is $\bar{\beta} > 0$ such that, for every $\beta \in (0, \bar{\beta}]$ and every bounded fee coefficient $\delta \geq 0$, uniformly for $0 \leq \nu \leq \bar{\nu}$,*

$$V_U(w) = V_0 + \alpha v^2 \lambda[F(q) - (1 + q)g(1)] + o(\alpha), \tag{14}$$

$$V_M(\alpha v^2 \delta, w) = V_0 + \alpha v^2\{\lambda F(q) - (1 + q)[\mathcal{E}_0(\lambda) + \delta]\} + o(\alpha). \tag{15}$$

Consequently,

$$\lim_{\alpha \downarrow 0} \frac{d^*_{\text{pkg}}(w)}{\alpha v^2} = \lambda g(1) - \mathcal{E}_0(\lambda). \tag{16}$$

A first-order optimal organization uses one protected direction converging to λ when $\lambda < t_0$. When $\lambda > t_0$, it combines a protected direction converging to t_0 with integrated direction one, with protected probability converging to $(1 - \lambda)/(1 - t_0)$.

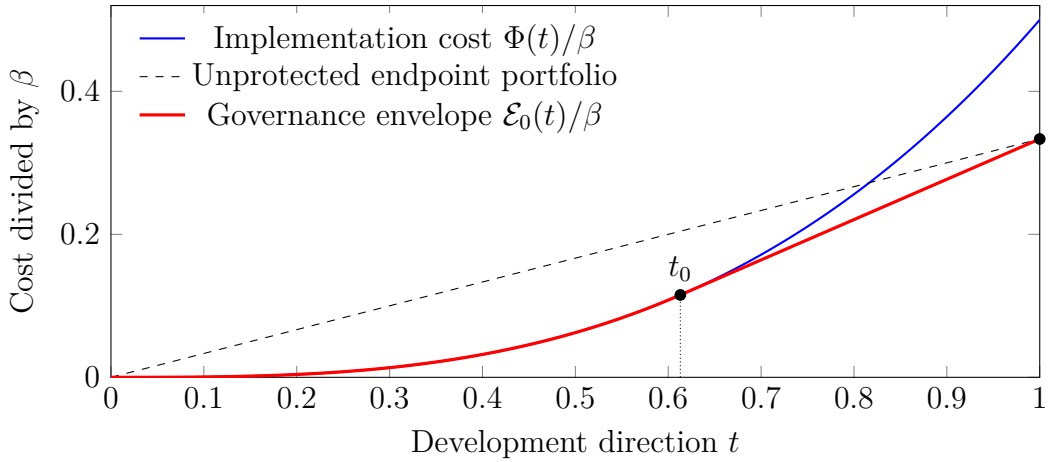


Figure 2: Delegation and integration under cubic development cost. Here $g(t) = \beta t^3/3$, so $\Phi = K$. The lower frontier follows protected implementation cost until its tangent reaches the integrated personal endpoint. The vertical distance from the endpoint-portfolio line to the frontier is the first-order package price divided by $\alpha v^2 \beta$.

The proof in Appendix B bounds the support value $S + \eta Q$ of every contract. Participation and a probability-growth bound control all schedules with nondecreasing personal claims. The financing cost of negative claim slopes controls schedules that use a more responsive incentive probability. A strict coverage inequality and the construction-cost optimum control the remaining choices. Integrated work and personal projects attain the unprotected support bound. Supporting lines to (13) then bound every financed lottery, and explicit protected contracts attain that bound to first order.

The factor $1 + q$ multiplies both the benefit of lower implementation cost and the fee in (15). Each unit of either cost reduces combined resources and also raises the amount that must be financed. At the margin, a unit of financing lets development move from work toward personal income, which yields q; the financing multiplier therefore converges to q, and each unit of cost is worth $1 + q$. The common factor cancels when solving for the fee. This explains why the first-order price is expressed directly as a difference between cost frontiers, although the financing problem determines the direction at which those frontiers are evaluated.

4.3 Wealth, technology, and organizational change

Let $D(\lambda) = \lambda g(1) - \mathcal{E}_0(\lambda)$ denote the normalized first-order price. On the single-developer segment,

$$D'(\lambda) = g(1) - \Phi'(\lambda). \tag{17}$$

On the mixed segment,

$$D(\lambda) = (1 - \lambda)[\Phi'(t_0) - g(1)], \tag{18}$$

which decreases strictly with wealth. Wealth raises the mean direction that financing permits. Whether consent then becomes more valuable depends on the marginal cost of implementing that direction relative to the integrated endpoint cost: the price rises while $\Phi'(\lambda) < g(1)$ and peaks at t_c, where $\Phi'(t_c) = g(1)$.

For power costs $g(t) = \beta t^n/n$, the peak is $t_c = [n^2 \max\{1/n, 1/2\}]^{-1/(n-1)}$. For $1 < n \leq 2$, $t_0 = 1$ and $D(\lambda) = \beta(\lambda - \lambda^n)/n$; the peak $t_c = n^{-1/(n-1)}$ is at most $1/2$, so the price falls with wealth throughout. For $n > 2$,

$$t_0^{n-1}[n - (n-1)t_0] = 2/n, \qquad D'(\lambda) = \beta\left[\frac{1}{n} - \frac{n}{2}\lambda^{n-1}\right]$$

in the single-developer region. Cubic cost gives a price that decreases with wealth throughout. Quintic cost gives an initially increasing price, followed by a decrease and then a mixed-organization segment (Figure 3). The price maximum and the organizational transition occur at different wealth levels. Greater convexity by itself does not order prices: it changes both development costs and the rents needed for implementation. Multiplying a fixed admissible cost function by a positive scalar multiplies the envelope and its price by the same scalar.

Cost exponent n	Organizational cutoff t_0	Wealth cutoff $t_0 - 1/2$	Price peak t_c
3/2	1	–	.4444
2	1	–	.5000
3	.6130	.1130	.4714
5	.6350	.1350	.5318
6	.6575	.1575	.5610

For $1 < n < 2$ and for cubic cost, the price peak lies below the relevant wealth domain; for quadratic cost it is at the zero-wealth boundary. The cutoff t_0 depends on cost shape, while cash feasibility also depends on income technology. For example, the $n = 3/2$ row is admissible under both fee conventions at $v = 1$, $r = 1/2$, $a = 3/2$, and $\pi = 24/25$, with a sufficiently small cost level. Online Appendix S.6 reports exact finite-α bounds for cubic, quintic, and sextic costs under three income technologies, including the transition to mixed governance.

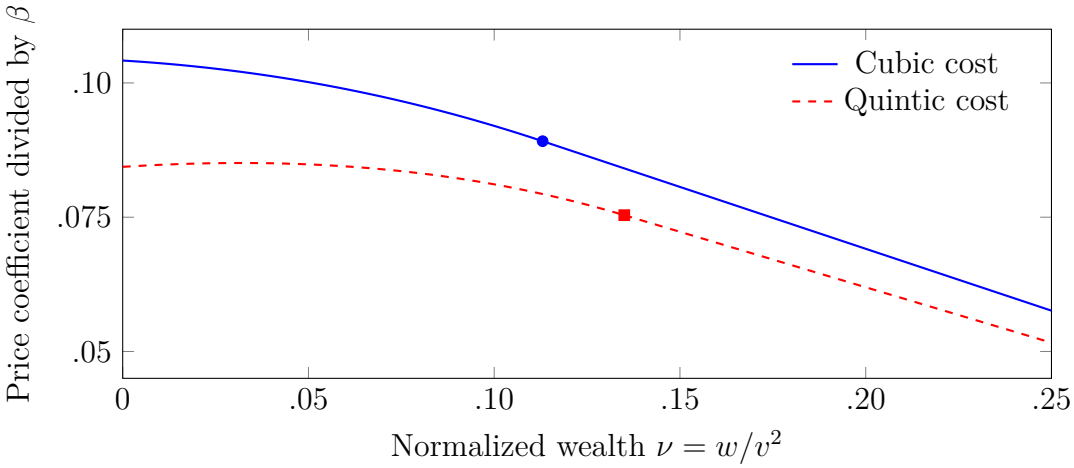


Figure 3: Technology and the wealth effect. Curves show $[\lambda g(1) - \mathcal{E}_0(\lambda)]/\beta$ for $g(t) = \beta t^n/n$. Quintic cost generates an initially increasing price. Markers identify the transition to a portfolio containing integrated personal development; after that transition each curve is linear and decreasing.

4.4 Strict incentives and unanimous amendments

Proposition 2 (Stable implementation of the first-order frontier)**.** *Under their respective assumptions and cost-level bounds, the values in Theorem 1, and those in Theorem 3 at fees strictly below its limiting cutoff, are attained up to $o(\alpha)$ by strictly implementing arrangements. For power costs, these arrangements can in addition be chosen renegotiation-proof in every realized branch. For general costs, the same holds when each contact direction satisfies the regularity conditions of Appendix C.*

Strict implementation follows from the margin (12). Renegotiation-proofness is obtained by attaining a support bound with strictly positive weights on every participant's continuation payoff. At positive-rent contacts, the proof selects contracts by maximizing creator value plus a positive, vanishing weight on D's rent. The rent is of order α, so this selection changes creator value by only $o(\alpha)$. The positive weight excludes amendments benefiting only D. At zero-rent contacts, a planner bound supplies the support weights directly.

Two kinds of contact direction arise. Where rents are positive, relevant protected support contracts converge to the flat-claim, state-01 arrangement, and their scaled developer payoffs converge to a function with a unique global maximum and strictly negative curvature. Where participation binds, D's limiting payoff has no strict maximum, since at zero rent direction zero also gives her zero. The selected contracts instead attain a planner bound that replaces developer compensation by development cost, and a contract paying a small rent implements the planner allocation strictly. Appendix C gives both arguments.

5 An exact quadratic benchmark

Quadratic cost has elasticity two, so $\Phi = g = K$: participation and incentive constraints coincide. It is the boundary between the zero-rent and positive-rent technologies of Theorem 1. Since $\Phi(1) = g(1)$, the organizational cutoff sits at the boundary, $t_0 = 1$, and the protected cost curve is itself the lower envelope up to the integrated personal endpoint. Quadratic cost also permits an exact solution at positive coverage and cost. Set

$$h(y) = \frac{\chi y^2}{2}, \qquad \alpha = \frac{\epsilon}{\gamma v}, \qquad \beta = \frac{\chi}{\alpha}, \qquad \varsigma = \sqrt{1+\chi}.$$

Consider the region

$$\begin{gathered} 3/10 \le q \le 1/2, \quad 0 < \alpha \le 1/4, \quad \beta_0(q,\alpha) < \beta \le 1/4, \quad a \ge r > 0, \\ \beta_0 = B(q) + \alpha B(q)^2/4, \qquad B(q) = \frac{q^2}{(1+2q)(1+q)}, \\ \kappa \ge \frac{(\pi-\epsilon)q}{4\gamma(1+q)}, \qquad 0 \le w/v^2 \le 1/4. \end{gathered} \tag{R}$$

The lower bound on β selects the region where the unprotected supporting multiplier is below q. The upper bound supplies cash for the zero-rent protected bonus. The coverage condition keeps additional coverage at zero in both solutions. The bounds on α and β give $\chi \le 1/16$; with the bounds on q, this keeps the unprotected personal claim x_U between $1/6$ and $1/4$, which makes the integrated personal endpoint uniquely optimal and cash feasible (Appendix D). The wealth bound gives $\lambda \le 3/4$.

Theorem 2 (Exact organizations and package value)**.** *Under* (R)*, let* $\eta_U \in (0, q)$ *solve*

$$1 + q + \frac{\alpha \eta_U^2}{2(1+2\eta_U)} = (1+\eta_U)\varsigma. \tag{19}$$

Set $x_U = \eta_U/(1+2\eta_U)$ *and* $H_U = \alpha x_U(1-x_U)$*. Then*

$$V_U(w) = \frac{v^2(1+\eta_U)}{2} + \eta_U w. \tag{20}$$

It is attained by a portfolio of two strictly implementing integrated projects, both with $\theta = 0$*:*

Project	*Capacity*	*Direction*	*Total claims* $(g_N, g_{10}, g_{01}, g_{11})$
Work	v	0	$(v, v, 0, v)$
Personal	v/ς	1	$(0, 0, x_U/\gamma, 0)$

The probability of the personal project is $\lambda/(1 - H_U/\varsigma)$.

For $0 \leq d < v^2/4$, *put* $\bar{\eta} = q/(1 - \alpha/4)$ *and define*

$$x(\eta) = \frac{\eta}{1 + 2\eta}, \qquad t(\eta) = \frac{q - \eta + \alpha\eta^2/[2(1 + 2\eta)]}{\chi(1 + \eta)}. \tag{21}$$

There is a unique interior solution $\eta_P \in (0, \bar{\eta})$ *to*

$$\frac{1}{2} - t_P - \frac{\chi}{2}t_P^2 + \alpha t_P x_P(1 - x_P) = \frac{d - w}{v^2}, \tag{22}$$

where $t_P = t(\eta_P)$ *and* $x_P = x(\eta_P)$. *The governance-package value is*

$$V_M(d, w) = v^2 \left(\frac{1}{2} + qt_P - \frac{\chi}{2}t_P^2 - \frac{\alpha}{2}t_P x_P^2\right) - d. \tag{23}$$

A single protected contract attains this value under weak IC, with $p = v$, $\theta = 0$, $b = 0$, $c = x_P/\gamma$, *and only a state-*01 *bonus. Its expected developer payment equals* $\chi v^2 t_P^2/2$. *The same value is the supremum under strict implementation and renegotiation-proofness.*

Equation (19) is the equality of the work and personal support bounds. Integration attains both, so richer bilateral amendment protocols cannot raise the unprotected value. The protected value follows from a planner relaxation that replaces developer compensation by development cost. A feasible zero-rent bonus attains the relaxation. That relaxation also bounds integration, so adding integrated branches to protected delegation does not improve the package frontier in this region. This establishes the exact benchmark for the expanded menu used in Theorem 1.

Zero rent and a flat claim leave the quadratic-cost developer indifferent across directions. A small positive claim slope and developer rent give strict implementation. At fee d, such a contract attains the planner value evaluated at $d+\varrho$, where $\varrho > 0$ is the developer rent; letting ϱ decrease to zero gives (23). The construction is also immune to unanimous amendments. Appendix D provides the argument.

Proposition 3 (Exact price and comparative statics)**.** *Under* (R), *there is a unique positive package threshold* $d^*(w)$. *Its adoption multiplier* $\widehat{\eta}_P$ *solves*

$$t(\widehat{\eta}_P)\left[1 + q - \alpha\left(\widehat{x}_P - \widehat{x}_P^2/2\right)\right] = (1 + \eta_U)\lambda, \tag{24}$$

where $\widehat{x}_P = x(\widehat{\eta}_P)$*, and* (22) *recovers the fee. Moreover,*

$$0 < d^*(w) \leq \frac{v^2(\varsigma - 1)}{2(\varsigma + 1)}, \tag{25}$$

$$\frac{d^*(w)}{v^2} = \frac{\alpha\beta}{2}\lambda(1-\lambda) + O(\alpha^2). \tag{26}$$

Along index paths staying in (R)*, the fee decreases with wealth and* q*, increases with* β *at fixed* α*, and decreases with* α *at fixed* χ*. It increases with* α *when* β *is fixed.*

The wealth derivative is $(\widehat{\eta}_P - \eta_U)/(1+\widehat{\eta}_P) < 0$. Wealth relaxes the unprotected financing problem more rapidly. The leading term in (26) does not depend on q, so the decline with q is a higher-order, finite-parameter effect. The coverage derivative holds development technology fixed; when coverage and costs increase proportionally, the cost effect reverses the sign. These comparative statics concern specified parameter paths, not an arbitrary change in personal revenue that simultaneously changes several indices.

Equivalently, the leading term of $d^*(w)$ in (26) is the Jensen gap

$$\lambda\frac{\chi v^2}{2} - \frac{\chi(\lambda v)^2}{2}.$$

At leading order, financing sets the intermediate direction and the personal-project portfolio weight equal to λ. Both face the same marginal financing price q; their difference is the convexity saving from developing one project at an intermediate direction. At $\lambda = 1/2$ this saving is $\chi v^2/8$. The extrapolated first-order expression vanishes at $\lambda = 1$, when wealth finances personal development directly; the finite-parameter theorem itself uses $\lambda \leq 3/4$.

5.1 Finite-parameter magnitudes

In (R), $\chi \leq 1/16$, and (25) gives $d^*/v^2 < 0.758\%$ and $d^*/V_U \leq 5/401 < 1.25\%$. The region selects relatively small covered personal income and development cost. Table 1 extends the comparison beyond this region, using exact rational certificates for feasible protected contracts, upper bounds on all unprotected lotteries, and a planner upper bound on the entire governance menu.

The effect also persists over a continuous set. With $v = r = a = 1$, $q \in [.30, .40]$, $\alpha \in [.8, 1.2]$, $\beta \in [.4, .5]$, $w \in [0, .02]$, and $\kappa \geq .15$, a strictly implementing protected contract finances a fee exceeding 4% of the optimized unprotected value at every point. The proof partitions the parameter rectangle into 548 rational boxes and bounds the same financing and incentive constraints used in the main model; the bounds are verified in exact

Table 1: Certified package-price intervals outside and inside the exact region

(q, α)	$\beta = .10$	$\beta = .25$	$\beta = .50$
$(.35, .25)$	$[.45, .46]$	$[1.14, 1.15]$	$[2.07, 6.30]$
$(.35, 1.00)$	$[1.77, 1.78]$	$[4.53, 4.54]$	$[7.04, 8.47]$
$(.35, 2.00)$	$[3.26, 3.27]$	$[7.47, 7.91]$	$[4.70, 5.91]$
$(.50, 1.00)$	$[1.54, 1.55]$	$[3.99, 4.00]$	$[7.37, 9.68]$

Entries are outward-rounded intervals for $100d^*/V_U$, with $v = r = a = 1$, $w = 0$, $\pi = 2(1+q)/3$, $\epsilon = \alpha/4$, $\chi = \alpha\beta$, and $\kappa = 1$. The first two cells lie in (R); they bracket exact values 0.459% and 1.150%. Lower bounds use strictly implementing contracts. Upper bounds use a common financing support bound and include organizational lotteries. Online Appendix S.4 gives the full table and certificate construction.

rational arithmetic with Bernstein coefficients (Online Appendix S.5). It provides a uniform magnitude result within the quadratic economy.

At very high development cost, both organizations rely on work and the value of consent falls. A general bound in the quadratic model is $d^*/V_U \leq q^2/\chi$: integrated work yields at least $v^2/2$, while total first-best resources are at most $v^2(1 + q^2/\chi)/2$. This explains why the price need not grow indefinitely with cost. The certified intervals establish model implications rather than a calibration of creator industries.

6 Paying for protection only when it is used

Suppose a fee is paid only in a branch that uses a protected developer. Let I_P identify such branches and define

$$V_M^{\mathrm{br}}(d, w) = \sup_{\mathcal{M}} \{\mathbb{E}[S - dI_P] : \mathbb{E}[Q - dI_P] \geq -w\}. \tag{27}$$

Because unprotected branches remain free, this value is always at least V_U. The economically relevant threshold is the highest fee per protected branch for which using protection strictly improves value.

For $d = \alpha v^2 \delta$, define

$$\mathcal{E}_\delta = \mathrm{co}\Big(\{(t, \Phi(t) + \delta) : 0 \leq t \leq 1\} \cup \{(0, 0), (1, g(1))\}\Big), \tag{28}$$

where co denotes the lower convex envelope. Charging protected projects shifts their curve upward while leaving integrated endpoints in place. Let t_c solve $\Phi'(t_c) = g(1)$. Branch fees can make protected directions near t_c relevant even outside the package contact set $\mathcal{A}$, so

the relevant set becomes

$$\mathcal{A}_{\mathrm{br}} = \Big[\min\{1/2, t_c\},\ \max\{\min\{\bar{\lambda}, t_0\}, t_c\}\Big].$$

Theorem 3 (The price per protected branch)**.** *Suppose Assumption 1 holds with $\mathcal{A}$ replaced by $\mathcal{A}_{\mathrm{br}}$. There is $\bar{\beta} > 0$ such that, for every $\beta \in (0, \bar{\beta}]$ and every bounded fee coefficient $\delta \geq 0$, uniformly for $0 \leq \nu \leq \bar{\nu}$,*

$$V_M^{\mathrm{br}}(\alpha v^2 \delta, w) = V_0 + \alpha v^2[\lambda F(q) - (1+q)\mathcal{E}_\delta(\lambda)] + o(\alpha). \tag{29}$$

Consequently,

$$\lim_{\alpha \downarrow 0} \frac{d^*_{\mathrm{br}}(w)}{\alpha v^2} = \delta_{\max} := t_c g(1) - \Phi(t_c) = \max_t \{t g(1) - \Phi(t)\}. \tag{30}$$

For $0 < \delta < \delta_{\max}$, unique cutoffs $t_L < t_c < t_R < t_0$ satisfy

$$t_L \Phi'(t_L) - \Phi(t_L) = \delta, \tag{31}$$

$$\Phi(t_R) + (1 - t_R)\Phi'(t_R) + \delta = g(1). \tag{32}$$

The first-order organization mixes work with protected direction t_L when $\lambda < t_L$, uses a single protected direction when $t_L \leq \lambda \leq t_R$, and mixes protected direction t_R with integrated personal development when $\lambda > t_R$.

The cutoffs merge at t_c as the fee approaches $\delta_{\max}$. A protected point lies below the unprotected endpoint line precisely when $\delta < \delta_{\max}$. Such a point improves the financed frontier at every interior mean direction by mixing it with the appropriate endpoint. Above the cutoff, all protected points are more expensive than the unprotected line. Appendix B strengthens this comparison using the exact unprotected support multiplier, which also rules out gains from a vanishing probability of protection.

For power costs, with $c_n = \max\{1/n, 1/2\}$,

$$t_c = (n^2 c_n)^{-1/(n-1)}, \qquad \delta_{\max} = \frac{\beta(n-1)t_c}{n^2}, \qquad t_L = \left[\frac{\delta}{\beta(n-1)c_n}\right]^{1/n}.$$

The branch-price coefficient is independent of wealth. The protected probability, and therefore expected protection spending, still changes with wealth. In contrast, the package-price coefficient satisfies

$$\lambda g(1) - \mathcal{E}_0(\lambda) \leq \delta_{\max}, \tag{33}$$

with equality only at $\lambda = t_c$ on the full interval $(0, 1)$. If t_c lies outside the wealth domain, equality is not reached there.

The distinction survives without developer rents. For quadratic cost, $\Phi = g$ and $t_c = 1/2$, and Theorems 1 and 3 apply directly. The package coefficient is $\beta\lambda(1-\lambda)/2$, while under branch fees

$$\delta_{\max} = \frac{\beta}{8}, \qquad t_L = \sqrt{2\delta/\beta}, \qquad t_R = 1 - \sqrt{2\delta/\beta}.$$

The fee convention also determines which protected directions must be feasible. With $v = r = a = 1$ and $q = .4$, the power cost with $n = 7/4$ satisfies (10) at both package and branch-fee contacts. With $n = 5/3$, the package condition holds on $[1/2, \bar{\lambda}]$, but the branch-fee cutoff $t_c \approx .465$ lies below $14/29 \approx .483$, the lowest direction at which the implementing personal claim fits in cash. A fixed charge for establishing a governance arrangement and a charge incurred only in projects that use it can therefore support different adoption and wealth patterns even with the same development technology.

7 Contract design and institutions

The results translate into guidance for artists, managers, and financiers. First, a management contract has two distinct terms that affect the artist: her payment share and her authority over reallocations of the manager's commission. The sale that consent blocks leaves the artist's payment share unchanged, so an artist who bargains only over shares leaves the value of consent unpriced. Second, the value of consent depends on the artist's financing needs and on how quickly development becomes costly. Under power costs that rise at most quadratically, the price of consent falls with wealth, so financially constrained artists who rely on outside financing gain most from a consent clause. Where costs rise faster, the price can rise with wealth at low wealth before falling. Third, consent does not make delegation optimal for every project. When the organizational cutoff t_0 lies below one, which requires manager rents, and the artist's wealth places λ above t_0, the best arrangement delegates balanced projects to a protected manager and places projects devoted entirely to artist-dependent income under the financier's direct control, where commitment is not needed. Fourth, the way protection is paid for matters. A charge levied on the whole arrangement depends on the artist's financing requirement; a charge levied only on projects that use a protected manager has a maximum independent of wealth, and wealth then affects only how often protection is used.

The right priced here concerns reallocations of the manager's commission. Assignment-consent and all-party amendment clauses provide institutional vehicles for such a right, and

existing management agreements allocate that authority in opposite ways. In the specimen agreement of the Featured Artists Coalition, Music Managers Forum, and Musicians' Union, the manager's commission is a share of the artist's income from recordings, compositions, live performance, brand partnerships, and other entertainment activities. The manager may not assign "this agreement or the benefit or burden of this agreement" without the artist's prior written consent, which "may be withheld at your absolute discretion." The clause separates the economic benefit of the agreement from the obligation to perform. The model studies the incentive consequences of that separation when the manager sells the contingent commission and remains responsible for development. The Desman–UOMO agreement instead lets the manager assign "this Agreement and the MANAGER's rights and obligations hereunder ... in whole or in part" to a third party, subject to notice and consultation. Both agreements also contain a key-person clause, which protects the artist if a named individual stops managing her career but does not reach a sale of the commission while that individual remains in charge, and both require joint signatures for amendments.[1]

The standard exclusive contract for singers issued by Korea's Ministry of Culture, Sports and Tourism contains several of the model's institutional elements in one document. The agency plans, produces, and schedules the artist's activities and the content built on them (Article 5(1)). Income from records and content and income from the artist's performing services are settled separately, each with its own agreed division between agency and artist (Article 12), consistent with the model's distinction between income that does not require the artist's later effort and income that does. The agency may transfer "all or part of its rights or status under this contract" to a third party only after obtaining the artist's prior written consent (Article 5(7)), and amendments require the written agreement of both parties (Article 15). An assignment of part of the agency's contractual income rights while it keeps its management duties is the separation of rights and duties studied here.[2]

All three contracts give the manager exclusive management of the artist's activities.[3] Ex-

[1]See the 2017 specimen agreement, Sections 1, 5.1, 10.1, 11.1, and 15.2, and the Desman–UOMO management agreement of June 23, 2009, Sections 10.01, 14.01, 15.03, and 15.05. The latter is filed with the SEC. Neither agreement refers to pledges of the commission. These documents identify contractual variation; they do not establish the incidence of incentive buyouts.

[2]Ministry of Culture, Sports and Tourism, Notice No. 2024-0021 (revised June 3, 2024), Standard Exclusive Contract for Popular Culture Artists (Singers), Articles 2(1), 5(1), 5(7), 12, and 15, available at law.go.kr. Translations are ours. The standard form is recommended, not mandatory.

[3]Specimen agreement, Section 1; Desman–UOMO agreement, Section 3.01, under which the manager is to "represent, supervise, advise, counsel and guide the ARTIST in the development and promotion of the ARTIST's career"; Korean standard contract, Article 2(1).

clusivity allocates career-level management, whereas integration in the model allocates the development decision for a particular project. Exclusive management of every project is consistent with the model whenever the optimal arrangement uses protected delegation alone: for power costs rising at most quadratically and, under faster-rising costs, below the organizational cutoff. Above that cutoff, the model favors placing the development of projects devoted entirely to artist-dependent income with the financier, outside the manager's remit. Such an arrangement requires project-specific carve-outs from an otherwise exclusive management mandate. The Korean standard contract permits the parties to reserve part of the management authority to the artist (Article 2(1)), which gives a contractual basis for allocating management rights across activities. The model identifies the incentive reallocations that protection must cover and the financing value it carries. Empirically, one should record whether the artist can block reallocations of the manager's commission, as distinct from assignment of the whole relationship and from key-person termination. A developer who continues to provide services can still change development direction when the payoff from doing so changes.

Syndicated credit agreements contain a comparable right. An assignment of a loan to a non-lender commonly requires the borrower's consent, not to be unreasonably withheld and deemed given absent timely objection; the requirement typically lapses while specified events of default continue.[4] The assignment leaves the borrower's repayment schedule unchanged, and borrowers that withhold consent have cited concern that a buyer would be less cooperative in later amendments or restructurings. The closest counterpart to the model is a lead arranger that continues to administer a loan while selling its retained share. Sufi (2007) finds that lead arrangers retain more of a loan when the borrower requires more monitoring, and Drucker and Puri (2009) find that loans that are sold carry more restrictive covenants. In these settings, as in the model, the anticipated sale of an incentive claim shapes the initial contract.

Film development and founder–engineer relationships can feature the same distinction between transferable project cash and income dependent on another productive party. As with the manager's commission, these applications require identifying who controls development and whose incentive exposure can be sold. A restriction that merely keeps a person employed need not preserve the exposure that governs the direction choice.

For measurement, payment shares and transfer authority should be recorded separately, since the artist's own share can remain unchanged when consent becomes valuable. Tech-

[4] Assignments to existing lenders, their affiliates, and approved funds are usually exempt. European leveraged loan documentation offers both consent and consultation-only alternatives.

nology matters through implementation cost as well as physical development cost: manager rents arise only where the elasticity of development cost exceeds two, and positive rents are necessary for the mixed-governance segment. A finding that the fee per protected project is insensitive to wealth can coexist with large changes in the use of protection.

The mixed segment predicts portfolios containing both integrated projects and protected delegated projects. Within the model, wealth raises the integrated-personal share after the organizational cutoff. This prediction concerns the joint organization of financing and development; it does not require an industry literally to choose projects by random draw. The commitment represented by the initial portfolio is that financing is evaluated across its projects before development decisions are made.

Intermediate information would introduce an additional role for amendments. An inefficiently rigid veto could then block a surplus-increasing response to new information. The present price measures the value of preserving incentives when those informational gains are absent. Extending the state space would require balancing that value against the forgone gains from adaptation. The continuous-region certificates establish magnitudes within the specified technology; how often such magnitudes arise depends on the distribution of technologies and financing needs.

8 Conclusion

A sale of a manager's commission can leave the artist's payment share unchanged while changing how her career is developed. Its availability restricts financing, and the artist can benefit from authority over that sale. The price of consent depends on the organizational alternative: integration avoids developer rents, while protected delegation preserves incentives for intermediate development. Their cost envelope determines both the value of the right and the transition to mixed governance. Implementation cost is the larger of development cost and an incentive bound. Quadratic costs sit at the boundary and isolate the convexity saving; where cost elasticity exceeds two, implementation rents appear and permit a reversal of the wealth effect. Charging protection at the arrangement level or only where it is used produces distinct prices from the same underlying technology.

A Accounts and the buyout bound

For each true cash state s, let $\widehat{s}(s)$ be A's retained substate and let $C(s)$ be actual cash. Replace payments by

$$T'_D(s) = T_D(\widehat{s}(s)), \qquad T'_F(s) = T_F(\widehat{s}(s)) + C(s) - C(\widehat{s}(s)).$$

This preserves A's retained income and D's payoff at every direction and returns discarded cash to F. A's retained income is weakly increasing in available cash, so the settlement rule implements full retention. For fixed $c = g_{11} - g_{10}$ and $b = g_{01} - c$, raise work claims to $g_N = v$ and $g_{10} = v + \min\{0, b\}$, with the increase assigned to F and returned through an initial advance. A's effort and all D deviations remain unchanged. This gives (3)–(4) for delegated contracts, branch by branch. Under integration the transformation can change the financier's development incentives; its accounts are used as a relaxation in the planner bound. The unprotected bound below is derived from the original schedule.

For the general buyout argument before this reduction, define retained incomes $u_{10} = v - g_{10}$, $u_{01} = R - g_{01}$, and

$$u_{11} = \max\{v + R - g_{11}, u_{10}, u_{01}, 0\}, \qquad c_E = R + u_{10} - u_{11}.$$

Let $\ell \geq 0$ be cash discarded in joint success. At direction u, the effective personal claim is $m(u) = (1-u)c_E + ug_{01}$. The personal-success intensity is

$$k(m) = r/R + \operatorname{clip}\Big(a(R-m)/R^2, 0, a/R\Big).$$

Write $G = (1-\omega)g_N + \omega g_{10}$. If f is F's continuation receipts net of fixed compensation, then

$$J(t) = pG(1-t) + p\omega t\{m(t)k(m(t)) - \ell(1-t)k(m(t))\} - h(pt) = f + R_D.$$

For $p > 0$, a complete purchase of D's claims uniquely induces $t = 0$; joint receipts become pG. Buyout stability therefore implies $J(t) \geq pG$. For $t > 0$, rearrangement gives

$$f \leq p\omega m(t)k(m(t)) - \frac{h(pt)}{t} - p\omega\ell(1-t)k(m(t)) - R_D. \tag{34}$$

At zero capacity the inequality is immediate. Under integration, the deviation to work-only development gives the same bound with $R_D = 0$.

Discarded cash enters $S + \eta Q$ with coefficient $-p\omega(1-t)k(m)(t+\eta) \leq 0$. Negative

effective claims contribute a nonpositive claim term. They can therefore be dropped in an upper bound. For positive-direction unprotected branches and any $\eta \geq 0$, completing the claim square yields

$$\begin{aligned} S + \eta Q \leq p\left[v + vqt - (1+\eta)\kappa\theta + \frac{\omega\eta^2}{2\gamma(2\eta + t)}\right] - \frac{1+\eta}{2}p^2 \\ - \left(1 + \frac{\eta}{t}\right) h(pt) - (1+\eta)R_D. \end{aligned} \tag{35}$$

This inequality is independent of quadratic development costs and holds for every branch of an unprotected portfolio.

B The full-menu first-order reduction

We prove Theorems 1 and 3. Bounds below are uniform on fixed compact wealth intervals with $\lambda \in (0,1)$ and bounded fee coefficients. Costs are evaluated on a common compact capacity set; quadratic construction costs supply the required bound.

B.1 Compensation and claim slopes

At a reduced schedule let $k(u) = 1 - \gamma \max\{c + bu, 0\}$. Normalized developer payments are (n, z_{10}, z, z_{11}). State probabilities are

$$((1-\omega)(1-u),\ \omega(1-u)(1-uk(u)),\ \omega u^2 k(u),\ \omega u(1-u)k(u)).$$

The work-state probabilities have nonpositive derivatives. In the affine effort region, the difference between the probability growth rates of states 01 and 11 is $1/[u(1-u)] > 0$. Let $W = C_D/(\alpha v^2)$ at $p = v$. Participation and nonnegative payments give $W \geq g(t)$. If $b \geq 0$, the largest positive growth rate is at most $2/t$, and a necessary first-order incentive condition gives $W \geq K(t)$. Hence $W \geq \Phi(t)$ whenever $b \geq 0$. Kinks are handled with the appropriate one-sided derivative; they cannot raise the maximum growth rate.

For $b < 0$, set $L = -\gamma b$. Cash feasibility gives $0 < L \leq E$ and $k \geq \Delta$, so the largest growth rate is at most $2/t + L/\Delta$. Combining participation with the first-order condition,

$$W \geq \max\left\{g(t), \frac{tg'(t)}{2 + Lt/\Delta}\right\} \geq \frac{\Phi(t)}{1 + Lt/(2\Delta)}. \tag{36}$$

The bounds include every nonnegative payment vector. They are necessary conditions; global implementation for the attaining construction was verified following (11).

At a support multiplier $\eta = q + O(\alpha)$, the loss of work claims from a negative slope, divided by $\alpha v^2(1+q)$, is $c_q L(1-t) + o(1)$, where $c_q = q/(1+q)$. The effective cost of a protected contract is therefore at least $W + c_q L(1-t) + o(1)$; extra coverage and deviations of the target claim only add to it (Appendix B.2).

For $a \in \mathcal{A}$ let $\ell_a(t) = \Phi(a) + \Phi'(a)(t-a)$ be the supporting line of the envelope at a. Since the derivative of $\Phi(a) + (1-a)\Phi'(a)$ in a is $(1-a)\Phi''(a) \geq 0$, (8) gives $\ell_a(1) \leq g(1)$. Let $a_m = \max \mathcal{A} < 1$. Strict convexity gives $D_1 = \Phi(1) - \ell_{a_m}(1) > 0$. Choose $0 < L_* \leq \min\{E, \Delta D_1/\Phi(1)\}$. Then $\Phi(1)/[1 + L_*/(2\Delta)] > \ell_{a_m}(1)$, and by continuity there is $T \in (a_m, 1)$ with

$$\frac{\Phi(t)}{1 + L_* t/(2\Delta)} > \ell_{a_m}(t) \qquad (T \leq t \leq 1). \tag{37}$$

The ratios $D_1/\Phi(1)$ and T do not depend on β.

Lemma B.1 (Negative claim slopes)**.** *Suppose*

$$\Phi(1) < 2\Delta c_q(1-T), \qquad g'(1) < c_q L_*. \tag{38}$$

Then $W + c_q L(1-t) \geq \ell_a(t)$ for every $L \in [0, E]$, every $t \in (0,1]$, and every $a \in \mathcal{A}$, strictly if $L > 0$ and $t < 1$.

Proof. If $L = 0$, (36) gives $W \geq \Phi(t) \geq \ell_a(t)$. If $t = 1$, participation gives $W \geq g(1) \geq \ell_a(1)$. It remains to consider $L > 0$ and $t < 1$. If $t \leq T$, (36) gives $W \geq \Phi(t) - \Phi(1)L/(2\Delta)$, while $c_q L(1-t) \geq c_q L(1-T) > \Phi(1)L/(2\Delta)$; hence the left side exceeds $\Phi(t) \geq \ell_a(t)$. If $t \geq T$ and $L \leq L_*$, (36) and (37) give $W > \ell_{a_m}(t) \geq \ell_a(t)$; the last inequality holds because the derivative of $\ell_a(t)$ in a is $\Phi''(a)(t-a) \geq 0$ for $t \geq a_m$. If $t \geq T$ and $L \geq L_*$, then $\ell_a(t) \leq \ell_a(1) \leq g(1)$, so participation and convexity give

$$\ell_a(t) - W \leq g(1) - g(t) \leq g'(1)(1-t) < c_q L(1-t). \qquad \square$$

When $\Phi = g$ on $[0,1]$, participation alone gives $W \geq \Phi(t) \geq \ell_a(t)$ for every slope, and (38) is not needed. Thus every relevant supporting line to the envelope is a valid lower bound on the effective cost of every protected contract. Adding a branch fee shifts that protected cost by δ. For branch fees, the same argument applies with $\mathcal{A}_{\mathrm{br}}$ in place of $\mathcal{A}$: its upper end does not exceed t_0, so $\ell_a(1) \leq g(1)$ continues to hold.

The cost level $\bar{\beta}$ in Theorems 1 and 3 can be any level below which three conditions hold: (38), unless $\Phi = g$ on $[0,1]$; the bonus cash condition $\Phi(t) < kt^2[x + k\psi_t(1-t)/t]$ on the relevant contact set; and $g'(1) < x(1-x)$ for the integrated endpoint in Appendix B.2. In each inequality one side is proportional to β and the other is independent of it.

B.2 Coverage, capacity, and the unprotected endpoints

Completing the claim square in the protected objective gives the optimal limiting claim $\gamma m = x = q/(1+2q)$. Deviations have a strictly negative quadratic contribution. The upper bound on the gain from coverage expansion is dominated by $(1+\eta)\kappa p\theta$ under Assumption 1. It follows that relevant first-order support maximizers have $\theta/\alpha \to 0$. In (35), the same assertion follows from $\eta^2/(2\eta+t) \leq \eta/2$ and the strict coverage condition. Bounds allowing reduced compensation are uniform in ω, so expanding coverage cannot avoid the claim-slope comparison.

At $\eta = q$, the zero-order capacity objective is $(1+q)(pv-p^2/2)$, with its unique maximum at $p = v$. On a common compact set the perturbation is uniform. Hence support maximizers converge to v, and the first-order support value is obtained by evaluating the perturbation there. Strict curvature in capacity justifies evaluating the first-order term at v.

For the positive-direction unprotected bound (35), set coverage to zero. The development term satisfies $0 \leq H(pt)/t \leq p \max_{[0,p]} H'$, so the bound equals its zero-order part $p[v+vqt]-(1+\eta)p^2/2$ up to an error of order α, uniformly in t. The zero-order part increases in t with slope pvq, so directions in $[0, t_1]$ fall short of direction one by a zero-order amount, for any fixed $t_1 < 1$. On $[t_1, 1]$, H is twice differentiable and the derivative of the bound with respect to t is $pvq + O(\alpha)$. Thus for sufficiently small α the positive-direction bound is maximized at $t = 1$. The work bound is attained at $t = 0$. Capacity optimization at the personal endpoint gives $p = v + O(\alpha)$, and the optimal claim is $\eta/[\gamma(1+2\eta)]$.

These positive-direction bounds are attainable by integration. At $p = v$ and $\eta = q$, choose total claims $(0, 0, x/\gamma, 0)$. Integrated profit, divided by αv^2, is $xu^2(1-xu) - g(u)$ up to a constant. Since $x < 1/2$,

$$x[(1-x) - u^2(1-xu)] = x(1-u)[1+u-x(1+u+u^2)] \geq x(1-x)(1-u),$$

while $g(1) - g(u) \leq g'(1)(1-u)$. The condition $g'(1) < x(1-x)$ therefore makes direction one the unique optimum with a margin proportional to $1-u$. Nearby capacities and claims retain strict implementation. The work schedule $(v, v, 0, v)$ strictly selects zero direction. Their limiting effective costs are therefore exactly zero and $g(1)$.

B.3 Financing duality and the multiplier

Let $\Psi(\eta)$ be the supremum of $S+\eta Q$ over a branch menu. With fees included in the branch accounts when appropriate, convexification and a strictly feasible work branch give

$$\sup\{\mathbb{E}S : \mathbb{E}Q \geq b\} = \inf_{\eta \geq 0}\{\Psi(\eta) - \eta b\}. \tag{39}$$

Two support branches suffice for a single financing constraint. Unbounded capacity cannot cause loss of compactness, since claims grow at most linearly with capacity and construction costs grow quadratically.

In a neighborhood of q, the zero-order support function is

$$\Psi_0(\eta) = \max\left\{\frac{v^2(1+\eta)}{2}, \frac{v^2(1+q)^2}{2(1+\eta)}\right\}.$$

Adding ηw gives left and right derivatives $v^2(\nu - 1/2) < 0$ and $v^2(\nu + 1/2) > 0$ at q. A uniform $O(\alpha)$ perturbation of this strict corner places any optimal multiplier at $\eta = q + O(\alpha)$. Write $\eta = q + \alpha e$. For a branch with effective cost $c(t)$, the first-order part of its support objective, apart from the common constant, is

$$tF(q) - (1+q)c(t) + e(1/2 - t).$$

If $a + st$ supports the cost envelope at λ, take $e = F(q) - (1+q)s$. The preceding branch bounds then give the same support inequality for every contract and hence every lottery. Substituting the financing requirement into (39) yields (14), (15), and (29). The envelopes are continuously differentiable, including their junctions, so the first-order multiplier is uniquely selected by their supporting slope.

B.4 Attainment and fee cutoffs

For the protected construction (11), direct accounts at $p = v$, $\theta = 0$ are

$$S/v^2 = 1/2 + qt - \alpha[tx^2/2 + \Phi(t)],$$
$$Q/v^2 = 1/2 - t + \alpha[tx(1-x) - \Phi(t)].$$

These accounts depend on the claims only through the target claim $\gamma m(t) = x$, which the claim slope leaves unchanged. Cash is strictly slack by (10) and the bonus cash condition.

The integrated personal endpoint has

$$S_1/v^2 = 1/2 + q - \alpha[x^2/2 + g(1)], \quad Q_1/v^2 = -1/2 + \alpha[x(1-x) - g(1)].$$

The work endpoint has $S_0/v^2 = Q_0/v^2 = 1/2$. Mixing the stated contact points of each supporting line and adjusting mixing probabilities by $O(\alpha)$ satisfies financing exactly; a single protected contract is combined with an $O(\alpha)$ probability of an endpoint, which leaves its incentive constraints unchanged. The perturbation (12) with $s \downarrow 0$ gives strict implementation at an $o(\alpha)$ cost, and strict cash slack persists. The identity

$$F(q) + x - x^2/2 = (1+q)x(1-x)$$

then gives the matching first-order values.

The package value is decreasing in its fee at least one for one. Comparing fixed fee coefficients below and above (16) brackets the adoption threshold. For branch fees below $\delta_{\max}$, one protected point lies strictly below the unprotected line; mixing with an endpoint improves value at every fixed $\lambda \in (0,1)$.

For a branch fee strictly above $\delta_{\max}$, use the *exact* unprotected supporting multiplier. The preceding argument gives

$$\eta_U = q + \alpha[F(q) - (1+q)g(1)] + o(\alpha).$$

Every protected branch, after its fee, has support value strictly below the unprotected maximum by at least

$$\alpha v^2(1+q)(\delta - \delta_{\max}) + o(\alpha),$$

uniformly over its contract choices. This excludes improvement even from protected probabilities tending to zero. The exact expanded value therefore equals V_U for sufficiently small α at any such fixed coefficient. Bracketing proves (30).

Finally, $\lambda g(1) - \mathcal{E}_0(\lambda)$ is concave. On its curved part it has derivative $g(1) - \Phi'(\lambda)$, zero only at $t_c < t_0$; on its straight part it decreases. Its unique maximum is $\delta_{\max}$, proving (33) and the equality claim.

B.5 Power costs and the strict margin

Let $g(t) = \beta t^n/n$ and put $r = u/t \geq 0$. If $n \geq 2$, then $\Phi = K = \beta t^n/2$ and $\psi_t = 0$. The difference between the two sides of (9) is

$$\frac{\beta t^n}{n}\left[r^n - 1 - \frac{n}{2}(r^2-1)\right] \geq 0,$$

by convexity of $s \mapsto s^{n/2}$ at $s = r^2 = 1$. If $1 < n < 2$, then $\Phi = g$ and $\psi_t = 2 - n$. The difference is

$$\frac{\beta t^n}{n} r^2\left[r^{n-2} - (3-n) + (2-n)r\right] \geq 0,$$

because the bracket compares the convex function r^{n-2} with its tangent at $r = 1$; at $r = 0$ the expression is zero by continuity. Thus (9) holds for every $n > 1$.

For the strict margin, write $\Phi_s = \Phi(t) + s$ and $\psi_s = 2 - tg'(t)/\Phi_s$. The bound on D's scaled payoff at deviation u is $\Phi_s r^2(1 + \psi_s - \psi_s r) - g(u)$, and $\Phi_s(1+\psi_s)$ and $\Phi_s\psi_s$ equal $3\Phi_s - tg'(t)$ and $2\Phi_s - tg'(t)$. The incentive margin at u is therefore affine in s with slope

$$1 - 3r^2 + 2r^3 = (1-r)^2(1+2r),$$

which gives (12). The cap on personal-success intensity only lowers deviation payoffs. For quadratic cost, $\psi_s = 2s/(\Phi(t)+s)$, and the construction coincides with that of Appendix D.

C Strict implementation and unanimous amendments

We give the supporting-contract construction for Proposition 2. Strict implementation alone follows from (12) and Appendix B; the construction here adds the amendment property. Amendments keep the realized capacity, coverage, and decision technology fixed and can replace cash-feasible payments and fixed compensation. Initial organizational mixing is chosen before that continuation stage.

C.1 Existence and a positive weight on the developer

For each small positive α, set $\tau_\alpha = \alpha$ and solve the auxiliary initial problem

$$\max \mathbb{E}[S - d_i + \tau_\alpha R_D] \quad \text{subject to} \quad \mathbb{E}[Q - d_i] \geq -w, \tag{40}$$

where $d_i = d$ for a package and $d_i = dI_P$ for branch fees. This is a selection device; the creator's objective remains (2) or (27). That objective puts no weight on D's rent and so

cannot by itself exclude an amendment that benefits D alone; the vanishing weight does.

Cash-bounded normalized payments, compact directions and coverage, and the quadratic construction cost restrict support maximization to a common compact payoff-relevant set. Nonnegative advances cannot escape this restriction: their cost has positive coefficient $1 + \eta - \tau_\alpha$. Weak IC is closed because continuation payoffs are continuous and the deviation set is compact. Thus the auxiliary support maximum exists. The work branch supplies strict financing feasibility, and convexification supplies a support multiplier and at most two active branches.

Near the limiting frontier, $R_D = O(\alpha)$. To see the relevant bound, compare with the attainable first-order support value. Giving D rent reduces $S+\eta Q$ by $(1+\eta)R_D$; compensation can earn only an $O(\alpha)$ improvement over the zero-order support optimum. In (40) its net coefficient remains bounded away from zero. Hence the auxiliary term changes optimized creator value by $O(\alpha^2)$. The dual argument of Appendix B still gives $\eta = q + O(\alpha)$ and the same first-order supporting slope.

Each active branch maximizes

$$S + \eta Q + \tau_\alpha R_D - (1 + \eta) d_i. \tag{41}$$

At the continuation stage construction and governance expenses are sunk. If Y_A denotes A's continuation payoff and f_F F's continuation receipts net of fixed compensation, (41) differs from

$$Y_A + (1 + \eta) f_F + \tau_\alpha R_D$$

only by a fixed constant. All participant weights are strictly positive. Any unanimous Pareto-improving amendment, including a randomized amendment within the same decision technology, would raise this support objective. This contradicts branch optimality.

C.2 Contact directions

Every supporting line of $\mathcal{E}_0$ or $\mathcal{E}_\delta$ touches the protected cost curve at most once, so each first-order organization has at most one protected contact direction a. For fixed interior wealth and fees below the branch cutoff, a is λ, t_0, t_L, or t_R, as appropriate, and lies in $\mathcal{A}$ or $\mathcal{A}_{\mathrm{br}}$. We call a *regular* if one of the following holds:

(R1) $\Phi(a) > g(a)$, and $U_0(u) = \Phi(a)(u/a)^2 - g(u)$ has a unique global maximum on $[0, 1]$ at a with $U_0''(a) < 0$;

(R2) $\Phi = g$ on a neighborhood of a, and (9) and (10), with x replaced by any nearby target claim, hold at nearby directions.

For power costs every contact is regular. If $n > 2$, then $\Phi = K > g$, the limit $U_0(u) = \beta[a^{n-2}u^2/2 - u^n/n]$ has a unique maximum at a, and $U_0''(a) = -\beta(n-2)a^{n-2}$. If $n \leq 2$, then $\Phi = g$ throughout, and the strict inequalities in Assumption 1 persist nearby. Proposition 2 for general costs assumes that the relevant contacts are regular.

C.3 Positive-rent contacts

Under (R1), the auxiliary support maxima above do not rely on an indifferent developer. Take any sequence of active protected support branches. At a positive-rent contact, a positive claim slope lowers the largest growth rate below $2/t$ and raises the compensation bound strictly above $K = \Phi$; a negative slope has a strict loss by Lemma B.1. These strict inequalities, the coverage bound, the strict claim-square penalty, and strict convexity of Φ imply

$$p \to v, \qquad \theta/\alpha \to 0, \qquad b \to 0, \qquad m \to x/\gamma, \qquad t \to a.$$

The cost bound's equality condition selects state 01 compensation. Work-state payments have nonpositive probability derivatives, and state 11 has a strictly lower probability growth rate than state 01 at any interior contact. Thus, in the normalized payment vector,

$$n/\alpha \to 0, \qquad z_{10} \to 0, \qquad z_{11} \to 0, \qquad z \to \frac{\Phi(a)}{\gamma(1-x)a^2} = \frac{g'(a)}{2\gamma(1-x)a}.$$

These conclusions follow from equality in the necessary compensation bound: retaining a positive first-order payment share in a strictly lower-growth state would cause a strictly positive support loss.

After division by αv^2, the developer's continuation payoff therefore converges uniformly on $[0,1]$ and in twice continuously differentiable form near a to U_0. By (R1), uniform convergence excludes all deviations outside a fixed neighborhood of a, and convergence of second derivatives makes the payoff strictly concave within that neighborhood for sufficiently small α. There is consequently a unique global maximizer, and the selected protected branch implements strictly.

C.4 Zero-rent contacts

Fix a contact a satisfying (R2). At zero rent, direction zero also gives D zero payoff, so strictness requires a perturbation. Define $F(\eta) = \eta^2/[2(1+2\eta)]$ and choose the unique multiplier near q solving

$$q - \eta_\alpha + \alpha F(\eta_\alpha) = \alpha(1+\eta_\alpha)g'(a). \tag{42}$$

The implicit function theorem gives $\eta_\alpha = q + \alpha[F(q) - (1+q)g'(a)] + O(\alpha^2)$. Since $\Phi = g$ near a, this is the first-order support slope required by the envelope.

Adding $(1+\eta)R_D$ to the support objective replaces compensation by physical development cost. Completing the claim square, dropping the nonpositive work-claim loss, and putting $y = pt$ gives

$$S + \eta Q + (1+\eta)R_D \le (1+\eta)(pv - p^2/2) + y\left[v(q-\eta) + \frac{\epsilon F(\eta)}{\gamma}\right] - (1+\eta)h_\alpha(y).$$

Here coverage is optimally zero: its coefficient is bounded above by $p[-(1+\eta)\kappa + (\pi - \epsilon)F(\eta)/\gamma] < 0$ for η near q. At $\eta = \eta_\alpha$, strict convexity of g and (42) make the unique unconstrained maximizers $p = v$ and $y = va$. They satisfy $0 < y < p$. The claim attaining the square is $m = \eta_\alpha/[\gamma(1+2\eta_\alpha)]$.

Use the construction of Section 4 at direction a and this claim. Condition (R2) supplies global IC and cash slack at the nearby claim. In the strict perturbation take $s = \varrho/(\alpha v^2)$, with $\varrho > 0$ and $\varrho = o(\alpha)$. The resulting contract has $C_D = h_\alpha(va) + \varrho$ and $R_D = \varrho$. Relative to the zero-rent allocation, both S and Q fall by exactly ϱ, so the displayed support bound remains an equality. Its continuation-payoff weights are 1 for A and $1 + \eta_\alpha$ for both F and D. They are strictly positive, which excludes every unanimous improvement.

This branch has the required first-order accounts under either fee convention. Combine it with the envelope's integrated branches and correct financing as in Appendix B: an $O(\alpha)$ adjustment of portfolio weights, adding an integrated endpoint if needed, meets the financing requirement exactly. Each branch is renegotiation-proof at its own positive continuation-payoff support; those weights need not be common across realized branches. The rent costs $o(\alpha)$, so the construction retains the first-order frontier. Appendix D gives the exact quadratic fee-shift version.

C.5 Unprotected branches

The unprotected active branches can be taken to be integrated endpoints. In the auxiliary problem, the rent term in the buyout support bound becomes $-(1+\eta-\tau_\alpha)R_D \le 0$. Dropping it gives a bound that, by Appendix B.2, is maximized for small α at the endpoints and attained there by strictly implementing integrated and work contracts. Unprotected delegated contracts attaining the same value need not be examined. Thus every active branch can be selected to implement strictly and to maximize a positive continuation-payoff support objective. The actual creator value differs from the weak-IC optimum by $o(\alpha)$.

D The finite-parameter benchmark

We provide the support argument, adoption equation, and stable strict approximation. The Online Appendix supplies the detailed parameter inequalities and additional compensation cases.

D.1 Unprotected support

For quadratic cost, maximizing (35) over capacity yields

$$S+\eta Q \le \frac{[N(t,\theta)]_+^2}{2A(t)}, \qquad \begin{cases} N = v + vqt - (1+\eta)\kappa\theta + \dfrac{\omega\eta^2}{2\gamma(2\eta+t)}, \\ A = 1+\eta+\chi(t^2+\eta t). \end{cases} \tag{43}$$

In (R), for $0 \le \eta \le q$, the bound increases in t and decreases in θ. To verify the direction claim, put $n = N(t,0)/v$; the inequalities

$$n' \ge 7q/8, \quad n \le 49/32, \quad A \ge 1, \quad A' \le 5q/8$$

give $2n'A - nA' \ge 203q/256 > 0$. The work and personal bounds coincide at (19). Its left-minus-right side decreases strictly. Its sign at q is negative precisely when $\beta > \beta_0(q,\alpha)$.

Both bounds equal $v^2(1+\eta_U)/2$. In the personal project the integrated objective is

$$J(u) = p_I \epsilon m_U u^2(1-\gamma m_U u) - \chi p_I^2 u^2/2, \quad p_I = v/\varsigma, \quad m_U = x_U/\gamma.$$

The inequalities $\eta_U \ge q - 3\chi/4 \ge 81/320$, $1/6 < x_U < 1/4$, and $\beta/\varsigma < x_U(2-3x_U)$ show that direction one is uniquely optimal and cash feasible. Its accounts are

$$Q_1/v^2 = H_U/\varsigma - 1/2, \quad S_1/v^2 = (1+q-\alpha x_U^2/2)/\varsigma - 1/2.$$

Together with the work branch, these attain the common support bound and the financing target, proving (20).

D.2 Protected support and the expanded menu

Developer participation gives $C_D \geq \chi p^2 t^2/2$. Substitution in the reduced accounts yields the planner support bound

$$\overline{\Psi}(\eta) = \max_{p,\theta,t,m} \left\{ p\left[(1+\eta)(v-\kappa\theta) + (vq-\eta v)t + \omega t\left(\eta m - \frac{\gamma}{2}(1+2\eta)m^2\right)\right] - \frac{1+\eta}{2}p^2(1+\chi t^2)\right\}. \tag{44}$$

The maximization can be restricted to $0 \leq m \leq R$. Integration also satisfies this relaxation with development cost replacing compensation. It therefore bounds all of $\mathcal{M}$, not only pure protected delegation.

For $\eta \in [0, \bar{\eta}]$, the optimal claim is $m = x(\eta)/\gamma < R$, and the coverage coefficient is negative under (R). Optimizing capacity leaves an objective proportional to

$$\frac{(1+\eta+Lt)^2}{1+\chi t^2}, \qquad L = q - \eta + \alpha\eta^2/[2(1+2\eta)].$$

Its interior optimum is (21) with $p = v$. The direction decreases strictly with η. The support derivative, equal to Q, increases strictly along this branch; its endpoints span financing levels from below $-7v^2/16$ to $v^2/2$. Thus (22) has a unique interior solution for the stated financing targets.

The flat-claim bonus is $z = \chi v/[2\epsilon(1-x_P)]$. It fits in state-01 cash precisely when $\beta \leq 2x_P(1-x_P)$. In (R) this inequality is strict; Online Appendix S.2 gives the exact cash boundary. Expected compensation equals development cost, so the contract attains (44) and proves (23) for the full menu.

D.3 Strict and renegotiation-proof approximation

Take the planner allocation at fee $d+\varrho$, for small $\varrho > 0$. Put $k = 1-\gamma m$ and $H_D = \chi p^2 t^2/2$, and choose

$$b = \frac{2k\varrho}{\gamma t(H_D+\varrho)}, \qquad c = m - bt, \qquad z = \frac{\chi p}{\epsilon(2k-\gamma bt)}.$$

Use total claims $(v, v, c+b, v+c)$ and D pay $(0, 0, z, 0)$. Cash slack ensures feasibility. Direct substitution gives $C_D = H_D + \varrho$ and

$$U_D(t) - U_D(u) = p\epsilon\gamma bz(u-t)^2(u+t/2) > 0 \quad (u \neq t).$$

Both financing and creator resources fall by ϱ relative to the planner allocation; at actual fee d the contract therefore attains $V_M(d+\varrho, w)$. The strengthened support bound

$$S + \eta Q + (1+\eta) R_D \leq \overline{\Psi}(\eta)$$

is attained. At the amendment stage it is a positive weighted sum of all continuation payoffs, so no unanimous improvement is possible. Taking $\varrho \downarrow 0$ gives the claimed supremum and establishes strict adoption at every fee below the threshold.

D.4 Adoption and derivatives

Write $D_U = \varsigma - H_U$. A feasible protected contract with $m = x_U/\gamma$ and $t_M = \lambda/D_U$ has the same creator continuation payoff as the unprotected optimum and a positive financing surplus:

$$\frac{Q_M + w}{v^2} = \frac{\chi\lambda}{2D_U^2}\left[\frac{2D_U}{\varsigma+1} - \lambda\right] > 0.$$

This proves strict gain at zero fee. At the common unprotected multiplier,

$$\overline{\Psi}(\eta_U) - \Psi_U(\eta_U) = \frac{v^2(1+\eta_U)\chi}{2(\varsigma+1)^2}.$$

Comparing both organizations at this multiplier yields (25). Strict decrease in the fee gives a unique crossing. Eliminating the financing requirement from equality of the two values gives (24).

The first-order expansion follows from

$$\eta_U = q + \alpha[F(q) - \beta(1+q)/2] + O(\alpha^2), \qquad \widehat{\eta}_P = q + O(\alpha).$$

Substitution in the adoption equation gives $t_P = \lambda\{1 + \alpha[x(1-x) - \beta/2]\} + O(\alpha^2)$ and then (26). For regularity, set $y = (q - \widehat{\eta}_P)/\alpha$. At $\alpha = 0$, the direction and adoption equations have Jacobian determinant $-(1+q)$ in (y, t); the implicit-function theorem supplies the expansion.

One obtains $\widehat{\eta}_P < \eta_U$ and $t_P < \lambda/D_U$ by evaluating (24) at η_U and using $2H_U > \varsigma - 1$.

Envelope differentiation gives

$$
\begin{aligned}
d_w^* &= \frac{\widehat{\eta}_P - \eta_U}{1+\widehat{\eta}_P} < 0, \\
d_q^* &= \frac{v^2}{1+\widehat{\eta}_P}(t_P - \lambda/D_U) < 0, \\
d_\beta^* &= \frac{\alpha v^2}{2}\left[\frac{1+\eta_U}{1+\widehat{\eta}_P}\frac{\lambda}{\varsigma D_U} - t_P^2\right] > 0, \\
d_\alpha^*|_\chi &= \frac{v^2}{1+\widehat{\eta}_P}\left[t_P F(\widehat{\eta}_P) - \frac{\lambda}{D_U}F(\eta_U)\right] < 0.
\end{aligned}
$$

The inequalities above, together with $\lambda < D_U/\varsigma$, give the displayed signs. Combining the last two derivatives proves the positive fixed-β coverage derivative; the bounding steps appear in Online Appendix S.3.

References

Aghion, Philippe, and Patrick Bolton. 1992. "An Incomplete Contracts Approach to Financial Contracting." *Review of Economic Studies* 59(3): 473–494.

Aghion, Philippe, Patrick Bolton, and Jean Tirole. 2004. "Exit Options in Corporate Finance: Liquidity versus Incentives." *Review of Finance* 8(3): 327–353.

Ayotte, Kenneth, and Patrick Bolton. 2011. "Optimal Property Rights in Financial Contracting." *Review of Financial Studies* 24(10): 3401–3433.

Bai, Chong-En, Zhigang Tao, and Changqi Wu. 2004. "Revenue Sharing and Control Rights in Team Production: Theories and Evidence from Joint Ventures." *RAND Journal of Economics* 35(2): 277–305.

Baliga, Sandeep, and Tomas Sjöström. 2009. "Contracting with Third Parties." *American Economic Journal: Microeconomics* 1(1): 75–100.

Bizer, David S., and Peter M. DeMarzo. 1992. "Sequential Banking." *Journal of Political Economy* 100(1): 41–61.

Demski, Joel S., and David E. M. Sappington. 1991. "Resolving Double Moral Hazard Problems with Buyout Agreements." *RAND Journal of Economics* 22(2): 232–240.

Dessí, Roberta. 2005. "Start-Up Finance, Monitoring, and Collusion." *RAND Journal of Economics* 36(2): 255–274.

Dewatripont, Mathias. 1989. "Renegotiation and Information Revelation over Time: The Case of Optimal Labor Contracts." *Quarterly Journal of Economics* 104(3): 589–619.

Drucker, Steven, and Manju Puri. 2009. "On Loan Sales, Loan Contracting, and Lending Relationships." *Review of Financial Studies* 22(7): 2635–2672.

Featured Artists Coalition, Music Managers Forum, and Musicians' Union. 2017. *Example Management Agreement*. Specimen contract.

Fudenberg, Drew, and Jean Tirole. 1990. "Moral Hazard and Renegotiation in Agency Contracts." *Econometrica* 58(6): 1279–1319.

Gorton, Gary B., and George G. Pennacchi. 1995. "Banks and Loan Sales: Marketing Nonmarketable Assets." *Journal of Monetary Economics* 35(3): 389–411.

Grossman, Sanford J., and Oliver D. Hart. 1986. "The Costs and Benefits of Ownership: A Theory of Vertical and Lateral Integration." *Journal of Political Economy* 94(4): 691–719.

Gryglewicz, Sebastian, Simon Mayer, and Erwan Morellec. 2024. "The Dynamics of Loan Sales and Lender Incentives." *Review of Financial Studies* 37(8): 2403–2460.

Hart, Oliver, and John Moore. 1988. "Incomplete Contracts and Renegotiation." *Econometrica* 56(4): 755–785.

Hart, Oliver, and John Moore. 1990. "Property Rights and the Nature of the Firm." *Journal of Political Economy* 98(6): 1119–1158.

Holmström, Bengt, and Paul Milgrom. 1991. "Multitask Principal–Agent Analyses: Incentive Contracts, Asset Ownership, and Job Design." *Journal of Law, Economics, & Organization* 7, Special Issue: 24–52.

Holmström, Bengt, and Jean Tirole. 1997. "Financial Intermediation, Loanable Funds, and the Real Sector." *Quarterly Journal of Economics* 112(3): 663–691.

Kahn, Charles M., and Dilip Mookherjee. 1998. "Competition and Incentives with Nonexclusive Contracts." *RAND Journal of Economics* 29(3): 443–465.

Laffont, Jean-Jacques, and David Martimort. 1998. "Collusion and Delegation." *RAND Journal of Economics* 29(2): 280–305.

Meyersson Milgrom, Eva M., Paul Milgrom, and Ravi Singh. 2023. "When Should Control Be Shared?" *Management Science* 69(1): 404–418.

Parlour, Christine A., and Guillaume Plantin. 2008. "Loan Sales and Relationship Banking." *Journal of Finance* 63(3): 1291–1314.

Pennacchi, George G. 1988. "Loan Sales and the Cost of Bank Capital." *Journal of Finance* 43(2): 375–396.

Segal, Ilya. 1999. "Contracting with Externalities." *Quarterly Journal of Economics* 114(2): 337–388.

Segal, Ilya, and Michael D. Whinston. 2003. "Robust Predictions for Bilateral Contracting with Externalities." *Econometrica* 71(3): 757–791.

Sufi, Amir. 2007. "Information Asymmetry and Financing Arrangements: Evidence from Syndicated Loans." *Journal of Finance* 62(2): 629–668.

Tirole, Jean. 1986. "Hierarchies and Bureaucracies: On the Role of Collusion in Organizations." *Journal of Law, Economics, & Organization* 2(2): 181–214.

Tirole, Jean. 2006. *The Theory of Corporate Finance*. Princeton University Press.

UOMO Media Inc. and Desman Inc. 2009. *Management Agreement*, June 23. SEC exhibit 10.18.

Vanasco, Victoria. 2017. "The Downside of Asset Screening for Market Liquidity." *Journal of Finance* 72(5): 1937–1982.